

KUMARESAN M

Software Engineer | Backend & Cloud Systems

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PROFESSIONAL SUMMARY

Software engineer with 2+ years of end-to-end ownership across backend systems, distributed cloud infrastructure, and IoT edge services in production. Architected a DICOM medical imaging pipeline on AWS (S3 + SQS + Lambda) — passed compliance audit, runs in live healthcare environments with zero incidents. Built Python-based IoT edge services on Raspberry Pi managing Bluetooth device pairing and FOTA firmware deployment. Shipped multi-tier RBAC APIs for a live e-commerce platform in Canada. Comfortable navigating ambiguous, complex systems from architecture to production — and collaborating closely with QA, product, and cross-functional stakeholders. Seeking an SDE-2 role at a product company to own high-impact backend systems.

CORE SKILLS

Backend	Node.js · Python · Express.js · FeathersJS · REST API Design · RBAC · WebSockets
Cloud & AWS	S3 · SQS · Lambda · Secrets Manager · Event-driven Architecture · Async Pipelines
Databases	PostgreSQL · MySQL · MongoDB · Firebase
Frontend	React.js · Next.js · Redux Toolkit · TypeScript · Tailwind CSS
Systems	Microservices · Distributed Systems · IoT / Edge · DICOM Imaging · FOTA (OTA)
Practices	Code Ownership · Structured Logging · Agile / Scrum · Swagger · Git · Linux

WORK EXPERIENCE

Full Stack Developer

Mar 2024 – Present

Rugas Technologies Pvt. Ltd. · Bangalore, KA

RADX.report — Medical Ultrasound Reporting Platform

- Owned end-to-end architecture and deployment of a production DICOM image ingestion pipeline — S3 + SQS + Lambda topology designed for asynchronous, retry-resilient medical image processing, with SQS chosen over direct Lambda invocation for back-pressure handling and exactly-once delivery semantics.
- Built and deployed Python IoT edge service on Raspberry Pi: Bluetooth device pairing with medical hardware, secure cloud sync, structured logging, and FOTA firmware deployment — enabling zero-touch remote updates across all production devices.
- Integrated FeathersJS backend with React.js frontend, enabling DICOM-based radiologist reporting workflows in production; managed credential lifecycle securely via AWS Secrets Manager.
- System passed remote healthcare compliance audit and has been running in production — demonstrating ownership from architecture decision through to production stability.

KarmaTickets.ca — Canada-based E-commerce Ticketing Platform

- Owned production backend and frontend features for a live Canada-based ticketing platform — authentication, role management, and order processing pipelines using Node.js, FeathersJS, and Next.js.
- Designed and shipped multi-tier RBAC system enforced at route middleware and data-access layer — improving access governance and eliminating privilege escalation vectors across admin, organiser, and buyer roles.
- Proactively partnered with QA and product stakeholders to resolve critical production issues beyond assigned scope — recognised with Certificate of Appreciation for ownership and responsiveness.

- ▶ Implemented structured logging and observability across backend and Linux device environments — reduced mean time to diagnose and resolve production incidents from hours to minutes.

KEY PROJECTS

DICOM Medical Imaging Pipeline

Production · 2024

AWS S3 · SQS · Lambda · Python · FeathersJS

- ▶ Owned end-to-end design of async ingestion pipeline: S3 upload → SQS event → Lambda processor → metadata storage. Architected with SQS over direct Lambda for retry resilience, back-pressure handling, and delivery guarantees across distributed components.
- ▶ System passed remote healthcare compliance audit. Running in production with zero manual intervention and zero critical incidents since deployment.

Raspberry Pi IoT Edge Service + FOTA

Production · 2024

Python · Bluetooth · Linux · Raspberry Pi

- ▶ Built Python background daemon managing Bluetooth device pairing, secure cloud communication, structured logging, and FOTA firmware deployment — zero-touch remote updates across production medical devices, no on-site intervention required.

MedCMS — Clinical Content Management System

Production · 2024

React · Node.js · PostgreSQL

- ▶ Delivered full-stack CMS for structured clinical data management — prioritised reliability, auditability, and feature completeness for regulated healthcare workflows. Reduced post-release defects through proactive QA collaboration.

EDUCATION

B.Tech, Textile Technology

Jul 2015 – May 2019

PSG College of Technology · Coimbatore, TN

Non-CS background. Self-directed transition into software engineering through production systems, DSA training in Java/Spring Boot, and hands-on AWS and IoT development.

CERTIFICATIONS & TRAINING

- ▶ **DSA in Java + Spring Boot Backend Specialisation** — Completion Certificate — Data structures, algorithms, Spring & Spring Boot

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